

Model Optimization and Tuning Phase Report

Date	21 June 2025
Team ID	SWTID1749707061
Project Title	Online Payments Fraud Detection Using Machine Learning
Maximum Marks	10 Marks

Model Optimization and Tuning Phase

The Model Optimization and Tuning Phase involves refining machine learning models for peak performance. It includes optimized model code, fine-tuning hyperparameters, comparing performance metrics, and justifying the final model selection for enhanced predictive accuracy and efficiency.

Hyperparameter Tuning Documentation (6 Marks):

Model	Tuned Hyperparameters	Optimal Values
Random Forest	<pre># Step 1: Define model rf_classifier = RandomForestClassifier() # Step 2: Define hyperparameters param_grid = { 'n_estimators': [100, 200], 'max_depth': [10, 20, None], 'min_samples_split': [2, 5, 10], 'min_samples_leaf': [1, 2, 4], 'class_weight': ['balanced'] }</pre>	<pre>accuracy_rf = accuracy_score(y_test, y_pred_rf) print(f"Optimal Hyperparameters: {grid_search.best_params_}") print(f"Accuracy on Test Set: {accuracy_rf}")</pre>

Decision Tree	<pre># Step 1: Define the Decision Tree classifier dt_classifier = DecisionTreeClassifier() # Step 2: Define the hyperparameters and their possible values for tuning param_grid = { 'criterion': ['gini', 'entropy'], 'splitter': ['best', 'random'], 'max_depth': [None, 10, 20, 30, 40, 50], 'min_samples_split': [2, 5, 10], 'min_samples_leaf': [1, 2, 4] }</pre>	<pre># Evaluate the performance of the tuned model accuracy = accuracy_score(y_test, y_pred) print(f"Optimal Hyperparameters: {grid_search.best_params_}") print(f"Accuracy on Test Set: {accuracy}")</pre>
Extra Tree	<pre># Define model et_classifier = ExtraTreesClassifier() # Define hyperparameters param_grid = { 'n_estimators': [100, 300], 'max_depth': [10, 25, None], 'min_samples_split': [2, 5], 'min_samples_leaf': [1, 2], 'class_weight': ['balanced'] }</pre>	<pre># Evaluate the performance of tuned model accuracy_et = accuracy_score(y_test, y_pred_et) print(f"Optimal Hyperparameters: {grid_search.best_params_}") print(f"Accuracy on Test Set: {accuracy_et}")</pre>
Support Vector Machine	<pre># Define model svc = SVC() # Define hyperparameters param_grid = { 'C': [1, 10], 'gamma': ['scale', 'auto'], 'kernel': ['rbf'], 'class_weight': ['balanced'] }</pre>	<pre># Evaluate the performance of tuned model accuracy_svc = accuracy_score(y_test, y_pred_svc) print(f"Optimal Hyperparameters: {grid_search.best_params_}") print(f"Accuracy on Test Set: {accuracy_svc}")</pre>
XG Boost	<pre># Define model xgb = XGBClassifier(use_label_encoder=False, eval_metric='logloss') # Define hyperparameters param_grid = { 'n_estimators': [100, 250], 'max_depth': [6, 8], 'learning_rate': [0.05, 0.1], 'subsample': [0.8], 'colsample_bytree': [0.8], 'scale_pos_weight': [50] }</pre>	<pre># Evaluate the performance of tuned model accuracy_xgb = accuracy_score(y_test, y_pred_xgb) print(f"Optimal Hyperparameters: {grid_search.best_params_}") print(f"Accuracy on Test Set: {accuracy_xgb}")</pre>

Performance Metrics Comparison Report (2 Marks):

Model	Optimized Metric																																				
Random Forest	Classification Report: <table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td><td>support</td></tr><tr><td>is Fraud</td><td>0.90</td><td>0.88</td><td>0.89</td><td>1587</td></tr><tr><td>is not Fraud</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1270934</td></tr><tr><td>accuracy</td><td></td><td></td><td>1.00</td><td>1272521</td></tr><tr><td>macro avg</td><td>0.95</td><td>0.94</td><td>0.94</td><td>1272521</td></tr><tr><td>weighted avg</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1272521</td></tr></table> Predicted is Fraud is not Fraud Actual <table><tr><td>is Fraud</td><td>1402</td><td>185</td></tr><tr><td>is not Fraud</td><td>117</td><td>1270817</td></tr></table>		precision	recall	f1-score	support	is Fraud	0.90	0.88	0.89	1587	is not Fraud	1.00	1.00	1.00	1270934	accuracy			1.00	1272521	macro avg	0.95	0.94	0.94	1272521	weighted avg	1.00	1.00	1.00	1272521	is Fraud	1402	185	is not Fraud	117	1270817
	precision	recall	f1-score	support																																	
is Fraud	0.90	0.88	0.89	1587																																	
is not Fraud	1.00	1.00	1.00	1270934																																	
accuracy			1.00	1272521																																	
macro avg	0.95	0.94	0.94	1272521																																	
weighted avg	1.00	1.00	1.00	1272521																																	
is Fraud	1402	185																																			
is not Fraud	117	1270817																																			
Decision Tree	Classification Report: <table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td><td>support</td></tr><tr><td>is Fraud</td><td>0.78</td><td>0.82</td><td>0.80</td><td>1587</td></tr><tr><td>is not Fraud</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1270934</td></tr><tr><td>accuracy</td><td></td><td></td><td>1.00</td><td>1272521</td></tr><tr><td>macro avg</td><td>0.89</td><td>0.91</td><td>0.90</td><td>1272521</td></tr><tr><td>weighted avg</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1272521</td></tr></table> Predicted is Fraud is not Fraud Actual <table><tr><td>is Fraud</td><td>1297</td><td>290</td></tr><tr><td>is not Fraud</td><td>681</td><td>1270253</td></tr></table>		precision	recall	f1-score	support	is Fraud	0.78	0.82	0.80	1587	is not Fraud	1.00	1.00	1.00	1270934	accuracy			1.00	1272521	macro avg	0.89	0.91	0.90	1272521	weighted avg	1.00	1.00	1.00	1272521	is Fraud	1297	290	is not Fraud	681	1270253
	precision	recall	f1-score	support																																	
is Fraud	0.78	0.82	0.80	1587																																	
is not Fraud	1.00	1.00	1.00	1270934																																	
accuracy			1.00	1272521																																	
macro avg	0.89	0.91	0.90	1272521																																	
weighted avg	1.00	1.00	1.00	1272521																																	
is Fraud	1297	290																																			
is not Fraud	681	1270253																																			
Extra Tree	Classification Report: <table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td><td>support</td></tr><tr><td>is Fraud</td><td>0.89</td><td>0.88</td><td>0.88</td><td>1587</td></tr><tr><td>is not Fraud</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1270934</td></tr><tr><td>accuracy</td><td></td><td></td><td>1.00</td><td>1272521</td></tr><tr><td>macro avg</td><td>0.94</td><td>0.94</td><td>0.94</td><td>1272521</td></tr><tr><td>weighted avg</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1272521</td></tr></table> Predicted is Fraud is not Fraud Actual <table><tr><td>is Fraud</td><td>1396</td><td>191</td></tr><tr><td>is not Fraud</td><td>185</td><td>1270749</td></tr></table>		precision	recall	f1-score	support	is Fraud	0.89	0.88	0.88	1587	is not Fraud	1.00	1.00	1.00	1270934	accuracy			1.00	1272521	macro avg	0.94	0.94	0.94	1272521	weighted avg	1.00	1.00	1.00	1272521	is Fraud	1396	191	is not Fraud	185	1270749
	precision	recall	f1-score	support																																	
is Fraud	0.89	0.88	0.88	1587																																	
is not Fraud	1.00	1.00	1.00	1270934																																	
accuracy			1.00	1272521																																	
macro avg	0.94	0.94	0.94	1272521																																	
weighted avg	1.00	1.00	1.00	1272521																																	
is Fraud	1396	191																																			
is not Fraud	185	1270749																																			
Support Vector Machine	Classification Report: <table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td><td>support</td></tr><tr><td>is Fraud</td><td>0.96</td><td>0.42</td><td>0.58</td><td>1587</td></tr><tr><td>is not Fraud</td><td>0.99</td><td>1.00</td><td>0.99</td><td>1270934</td></tr><tr><td>accuracy</td><td></td><td></td><td>0.99</td><td>1272521</td></tr><tr><td>macro avg</td><td>0.97</td><td>0.71</td><td>0.78</td><td>1272521</td></tr><tr><td>weighted avg</td><td>0.99</td><td>0.99</td><td>0.99</td><td>1272521</td></tr></table> Predicted is Fraud is not Fraud Actual <table><tr><td>is Fraud</td><td>669</td><td>918</td></tr><tr><td>is not Fraud</td><td>81</td><td>1270853</td></tr></table>		precision	recall	f1-score	support	is Fraud	0.96	0.42	0.58	1587	is not Fraud	0.99	1.00	0.99	1270934	accuracy			0.99	1272521	macro avg	0.97	0.71	0.78	1272521	weighted avg	0.99	0.99	0.99	1272521	is Fraud	669	918	is not Fraud	81	1270853
	precision	recall	f1-score	support																																	
is Fraud	0.96	0.42	0.58	1587																																	
is not Fraud	0.99	1.00	0.99	1270934																																	
accuracy			0.99	1272521																																	
macro avg	0.97	0.71	0.78	1272521																																	
weighted avg	0.99	0.99	0.99	1272521																																	
is Fraud	669	918																																			
is not Fraud	81	1270853																																			
XG Boost	Classification Report: <table><tr><td></td><td>precision</td><td>recall</td><td>f1-score</td><td>support</td></tr><tr><td>is Fraud</td><td>0.87</td><td>0.88</td><td>0.88</td><td>1587</td></tr><tr><td>is not Fraud</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1270934</td></tr><tr><td>accuracy</td><td></td><td></td><td>1.00</td><td>1272521</td></tr><tr><td>macro avg</td><td>0.94</td><td>0.94</td><td>0.94</td><td>1272521</td></tr><tr><td>weighted avg</td><td>1.00</td><td>1.00</td><td>1.00</td><td>1272521</td></tr></table> Predicted is Fraud is not Fraud Actual <table><tr><td>is Fraud</td><td>1398</td><td>189</td></tr><tr><td>is not Fraud</td><td>215</td><td>1270719</td></tr></table>		precision	recall	f1-score	support	is Fraud	0.87	0.88	0.88	1587	is not Fraud	1.00	1.00	1.00	1270934	accuracy			1.00	1272521	macro avg	0.94	0.94	0.94	1272521	weighted avg	1.00	1.00	1.00	1272521	is Fraud	1398	189	is not Fraud	215	1270719
	precision	recall	f1-score	support																																	
is Fraud	0.87	0.88	0.88	1587																																	
is not Fraud	1.00	1.00	1.00	1270934																																	
accuracy			1.00	1272521																																	
macro avg	0.94	0.94	0.94	1272521																																	
weighted avg	1.00	1.00	1.00	1272521																																	
is Fraud	1398	189																																			
is not Fraud	215	1270719																																			

Final Model Selection Justification (2 Marks):

Final Model	Reasoning
Random Forest Classifier	Random Forest Classifier is selected as the final model due to its superior performance in detecting fraud transactions with high recall, precision, and F1-score, while maintaining extremely low false negative rates.